

LAB #1

The Internet Protocols

CS471 – Web Technologies

 **Dr. Sumayah alkhalf**

Shaden Alkhalifah – 431202284

Part 1: Capturing and Analyzing HTTP Traffic

The image displays a Wireshark packet capture of an HTTP GET request. The top pane shows the packet list with 8 packets. The middle pane shows the packet details for packet 1311, highlighting the Hypertext Transfer Protocol section. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1311	47.904186900	192.168.0.168	34.223.124.45	HTTP	481	GET / HTTP/1.1
1317	48.893196200	34.223.124.45	192.168.0.168	HTTP	1027	HTTP/1.1 200 OK (text/html)
1461	51.985007600	192.168.0.168	34.223.124.45	HTTP	540	GET /online HTTP/1.1
1466	52.232327200	34.223.124.45	192.168.0.168	HTTP	631	HTTP/1.1 301 Moved Permanently (text/html)
1469	52.240391300	192.168.0.168	34.223.124.45	HTTP	541	GET /online/ HTTP/1.1
1477	52.485678400	34.223.124.45	192.168.0.168	HTTP	273	HTTP/1.1 200 OK (text/html)
1480	52.551758300	192.168.0.168	34.223.124.45	HTTP	473	GET /favicon.ico HTTP/1.1
1484	52.814988200	34.223.124.45	192.168.0.168	HTTP	470	HTTP/1.1 200 OK (PNG)

Packet 1311 details:

- Ethernet II, Src: Intel_c4:a3:b0 (28:95:29:c4:a3:b0), Dst: TplinkT
- Internet Protocol Version 4, Src: 192.168.0.168, Dst: 34.223.124.4
- Transmission Control Protocol, Src Port: 54922, Dst Port: 80, Seq: 3372
- Hypertext Transfer Protocol

Raw packet data (hex and ASCII):

```

0000  50 d4 f7 5a 0c 17 28 95 29 c4 a3 b0 08 00 45 00  P...N...
0010  01 d3 4a ae 40 00 80 06 00 00 c0 a8 00 a8 22 df  .N...Ho...
0020  7c 2d 46 8a 00 50 a2 df 61 ce b9 89 76 77 50 18  |...P...a...
0030  ff ff 62 22 00 00 47 54 20 2f 20 48 54 5a 59 58  |"..."GE T...
0040  2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6e 65 76 65  /1.1..Ho...
0050  72 73 73 6c 2e 63 6f 6d 0d 04 c3 6f 6e 6e 65 63  rssl.com...
0060  74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65  tion: ke...
0070  0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75  ..Upgrad...
0080  72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d 0a  re-Req...
0090  55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69  User-Age...
00a0  6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73  lla/5.0...
00b0  20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b  NT 10.0...
00c0  20 78 36 34 29 20 41 70 70 6c 65 57 65 62 4b 69  x64) Ap...
00d0  74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c  t/537.3...
00e0  20 6c 69 6b 65 20 47 65 63 6b 6f 29 20 43 68 72  like Ge...
00f0  6f 6d 65 2f 31 35 32 2e 30 2e 30 2e 30 20 53 61  ome/152...
0100  66 61 72 69 2f 35 33 37 2e 33 36 0d 0a 41 63 63  fari/537...
0110  65 70 74 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61  ept: tex...
0120  70 70 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c  pplicati...

```

Part 2 - Task 1: Follow TCP Stream

Wireshark · Follow TCP Stream (tcp.stream eq 66) · شبكة Wi-Fi

GET / HTTP/1.1
Host: neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9

HTTP/1.1 200 OK
Date: Fri, 11 Sep 2026 07:10:20 GMT
Server: Apache/2.4.66 ()
Upgrade: h2,h2c
Connection: Upgrade, Keep-Alive
Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT
ETag: "f79-5e28b29d38e93-gzip"
Accept-Ranges: bytes
Vary: Accept-Encoding
Content-Encoding: gzip
Content-Length: 1900
Keep-Alive: timeout=5, max=100
Content-Type: text/html; charset=UTF-8

.....Wkn....-B#...C.Z.k.R...q..."...A...&.....Q...!...+..yr.\!_u.PC%.<....zW}
UZ...7.N.....'...7...{.V...hJ...12..1).2Z...~.C.Vk[...#. 'k.t#*...+.....0.R.{.T.&\$.zB^?.
...9Pz.E^...X.[m...=.*.Q%.V.+...T.2..._xf/e5 1
...y..s..#.....?..r..v..e%{U...^v..
.p@...q..kQ...T...

Packet 1316. 1 client pkt, 1 server pkts, 1 turn. Click to select.

Entire conversation (2700 bytes) Show as ASCII No delta times Stream 11

Find: Case sensitive Find Next

Filter Out This Stream Print Save as... Back Close Help

Part 2 - Task 2: TCP Handshake & Data Transfer

No.	Time	Source	Destination	Protocol	Length	Info
1266	46.248887600	192.168.0.1	192.168.0.168	TCP	56	53 → 64212 [FIN, ACK] Seq=52 Ack=37 Win=14600 Len=0
1267	46.249066200	192.168.0.168	192.168.0.1	TCP	54	53075 → 53 [ACK] Seq=37 Ack=53 Win=65280 Len=0
1268	46.249208900	192.168.0.168	192.168.0.1	TCP	54	65091 → 53 [ACK] Seq=37 Ack=94 Win=65280 Len=0
1269	46.249298500	192.168.0.168	192.168.0.1	TCP	54	64212 → 53 [ACK] Seq=37 Ack=53 Win=65280 Len=0
1306	47.062519600	192.168.0.168	34.223.124.45	TCP	66	[TCP Retransmission] 51164 → 80 [SYN] Seq=0 Win=65535
1307	47.062746100	192.168.0.168	34.223.124.45	TCP	66	[TCP Retransmission] 59768 → 80 [SYN] Seq=0 Win=65535
1308	47.270631700	192.168.0.168	34.223.124.45	TCP	66	[TCP Retransmission] 54922 → 80 [SYN] Seq=0 Win=65535
1309	47.903532700	34.223.124.45	192.168.0.168	TCP	58	80 → 54922 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS
1310	47.903758700	192.168.0.168	34.223.124.45	TCP	54	54922 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0
1311	47.904186900	192.168.0.168	34.223.124.45	HTTP	481	GET / HTTP/1.1
1312	47.919514200	52.110.18.10	192.168.0.168	TLSv1.2	92	Application Data[Malformed Packet]
1313	47.962191200	192.168.0.168	52.110.18.10	TCP	54	52287 → 443 [ACK] Seq=79 Ack=141 Win=253 Len=0
1314	48.229179000	34.223.124.45	192.168.0.168	TCP	56	80 → 54922 [ACK] Seq=1 Ack=428 Win=26800 Len=0
1316	48.893185000	34.223.124.45	192.168.0.168	TCP	1354	80 → 54922 [ACK] Seq=1 Ack=428 Win=26800 Len=1300 [TC
1317	48.893196200	34.223.124.45	192.168.0.168	HTTP	1027	HTTP/1.1 200 OK (text/html)
1318	48.893303000	192.168.0.168	34.223.124.45	TCP	54	54922 → 80 [ACK] Seq=428 Ack=2274 Win=65535 Len=0
1319	48.936124100	192.168.0.168	192.168.0.1	TCP	66	63534 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=25
1320	48.937143100	192.168.0.168	192.168.0.1	TCP	66	49879 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=25
1321	48.948715600	192.168.0.1	192.168.0.168	TCP	68	53 → 63534 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS
1322	48.948717800	192.168.0.1	192.168.0.168	TCP	68	53 → 49879 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS
1323	48.949340400	192.168.0.168	192.168.0.1	TCP	54	63534 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
1324	48.949507100	192.168.0.168	192.168.0.1	TCP	54	49879 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
1325	48.949697700	192.168.0.168	192.168.0.1	TCP	56	49879 → 53 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=2 [TC
1326	48.949844500	192.168.0.168	192.168.0.1	DNS	106	Standard query 2058 A google.com. flags=0x00000000

Part 3: Capturing and Analyzing UDP Traffic

No.	Time	Source	Destination	Protocol	Length	Info
4737	59.325753700	216.239.38.120	192.168.0.168	QUIC	66	Protected Payload (KPo)
4738	59.378184600	142.251.151.4	192.168.0.168	QUIC	72	Protected Payload (KPo)
4739	59.388752400	142.251.151.4	192.168.0.168	QUIC	221	Protected Payload (KPo)
4740	59.388753600	142.251.151.4	192.168.0.168	QUIC	64	Protected Payload (KPo)
4741	59.411977600	192.168.0.168	142.251.151.4	QUIC	79	Protected Payload (KPo), DCID=f962203c60a4f566
4742	59.485558300	173.194.141.104	192.168.0.168	QUIC	573	Protected Payload (KPo)
4743	59.485560700	173.194.141.104	192.168.0.168	QUIC	159	Protected Payload (KPo)
4744	59.485561600	173.194.141.104	192.168.0.168	QUIC	65	Protected Payload (KPo)
4745	59.485562100	173.194.141.104	192.168.0.168	QUIC	70	Protected Payload (KPo)
4746	59.485564600	173.194.141.104	192.168.0.168	QUIC	482	Protected Payload (KPo)
4747	59.485565600	173.194.141.104	192.168.0.168	QUIC	84	Protected Payload (KPo)
4748	59.486167600	192.168.0.168	173.194.141.104	QUIC	74	Protected Payload (KPo), DCID=efd2d32aaa87b913
4749	59.486498700	192.168.0.168	173.194.141.104	QUIC	77	Protected Payload (KPo), DCID=efd2d32aaa87b913
4754	59.503117100	142.251.151.4	192.168.0.168	QUIC	68	Protected Payload (KPo)
4761	59.515754300	192.168.0.168	173.194.141.104	QUIC	78	Protected Payload (KPo), DCID=efd2d32aaa87b913
4774	59.613138300	192.168.0.168	172.217.20.97	QUIC	1292	Initial, DCID=9ad9abc45c987376, PKN: 1, CRYPTO, CRYPTO, PADDING, CRYPTO, PING, PADDING, PING...
4775	59.613290600	192.168.0.168	172.217.20.97	QUIC	1292	Initial, DCID=9ad9abc45c987376, PKN: 2, PADDING, PING, PING, CRYPTO, PING, PADDING, CRYPTO...
4781	59.606215300	172.217.20.97	192.168.0.168	QUIC	83	Initial, SCID=fad9abc45c987376, PKN: 1, ACK

▶ Frame 4749: Packet, 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface 0
 ▶ Ethernet II, Src: Intel_C4:a3:b0 (28:95:29:c4:a3:b0), Dst: TplinkTechno_5a:0c:17 (58:9f:9d:5a:0c:17)
 ▶ Internet Protocol Version 4, Src: 192.168.0.168, Dst: 173.194.141.104
 ▶ User Datagram Protocol, Src Port: 63285, Dst Port: 443
 Source Port: 63285
 Destination Port: 443
 Length: 43
 Checksum: 0xfc7 [unverified]
 [Checksum Status: Unverified]
 [Stream index: 29]
 [Stream Packet Number: 26]
 [Timestamps]
 UDP payload (35 bytes)
 QUIC IETF

Source Port (udp.srcport): 1 bytes
 Packets: 13360 - Displayed: 12171 (91.1%) - Dropped: 0 (0.0%) - Profile: Default

Task 1: Fill in the following table and provide reasons.

Feature	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP uses a 3-way handshake and guarantees that every packet arrives. UDP has no connection and no delivery guarantee.
Data Integrity and Ordering	TCP	TCP numbers packets so they arrive in order without errors. UDP does not order packets.

Task 2: Identify the use Cases and Performance of TCP and UDP.

Feature	TCP	UDP
Use cases	Websites, downloading files, emails.	Live streams, video calls, online games.
Performance	Slower (checks for errors and lost data).	Faster (sends data directly with no delays).



[Access the GitHub Repository](#)